



Product Requirements Document

Version 0.1 · May 11, 2026 · Status: Approved — Ready for Design

1. Overview

Tithika is a mobile application for iOS and Android that calculates and displays **tithis** (lunar phases per the Hindu calendar) with astronomical accuracy. The app focuses on the tithi and tithi-derived festivals and observances rather than the full Panchanga.

Audience: General users — devout users who need accuracy for fasts and observances, and curious users alike. The app must be approachable while the underlying mathematics must be correct.

License: Open source.

2. Platforms & Tech Stack

Item	Choice
Target platforms	iOS and Android
Framework	Flutter (single codebase)
Language	Dart
Astronomical engine	Swiss Ephemeris (Astrodienst libswe)
Native integration	Dart FFI (dart:ffi) — evaluate the sweph package first; fall back to custom FFI bindings if needed
Ephemeris data	Bundled as Flutter assets (~10 MB, Sun + Moon files)
Operation mode	Fully offline — no backend required
Swiss Ephemeris license	AGPL — open-source app, no commercial license needed

3. Calendar System

Item	Default	Configurable?
Calculation method	Drik (true astronomical positions)	Future: add Vakya
Month system	Purnimanta — month ends on Purnima (North Indian convention)	Yes — Amanta toggle in Settings
Day-boundary (grid)	Sunrise-prevailing tithi	Not configurable
Day-boundary (main view)	Currently-active tithi + 'transitions to X at HH:MM'	Not configurable

4. Core Calculations

Per location and date, the app computes (all calculations use user's lat/lon and timezone):

- Sun longitude & Moon longitude — from Swiss Ephemeris.
- Tithi — $((\text{moonLong} - \text{sunLong}) \bmod 360) / 12^\circ$, value 1-30. 1-15 = Shukla Paksha (waxing); 16-30 = Krishna Paksha (waning).
- Tithi start & end times — when the longitude difference crosses 12° boundaries.
- Nakshatra — $\text{moonLong} / (360/27)$, one of 27 nakshatras, with end time.
- Sunrise & sunset — via `swe_rise_trans`; sunrise also drives the day-boundary rule.
- Lunar month — Chaitra through Phalgun (Purnimanta convention).
- Sun's zodiacal sign — Makara, Mesha, etc. — needed for solar festival detection.

5. UI / Screen Structure

5.1 Main Screen (Default View)

Top 75% — Today / selected date:

- Prominent Gregorian date (large, hero element)
- Hindu month + paksha line (e.g., *Vaishakha · Shukla Paksha*)
- Moon image — rendered programmatically from current illumination percentage (derived from tithi)
- Tithi name and number (e.g., *Shukla Pratipada*)
- Current tithi start time and end time (e.g., 6:42 AM – 8:17 PM)
- Nakshatra + end time
- Sunrise & sunset times
- Festival name if applicable · Special-day indicator if applicable

Bottom 25% — 4-day strip: Previous day | **Current day (highlighted)** | Next day | Day after. Each cell shows date, tithi number, and paksha color. Tapping a cell selects that date.

5.2 Month View

Toggleable from the main screen. Standard 7-column month grid. Each cell displays:

- Gregorian date
- Tithi number (sunrise-prevailing)
- Small programmatic moon-phase icon (accurate per tithi)
- Paksha color — Shukla = light/gold, Krishna = dark/blue
- Indicator dot for special tithi or festival days

5.3 First-Launch Onboarding

On first open the user is prompted to set their location before reaching the main screen. Two options: **Use my current location** (GPS, resolves to nearest city name) or **Enter manually** (searchable city picker).

Fallback: If the user dismisses without providing a location, the app uses **Ujjain** (75.78°E, 23.18°N — classical Hindu calendrical reference meridian) as a temporary location. A persistent banner reads "Tap to set your location" until the user provides one.

5.4 Settings Screen

Setting	Options	Default
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Location	GPS (refresh from current) / Manual city picker	Set during onboarding
Language	English / Hindi (Latin script) / Hindi (Devanagari)	English
Month system	Purnimanta / Amanta	Purnimanta

6. Special Tithi Indicators

Subtle visual indicators appear on the calendar grid and as a callout on the main view:

Tithi	When	Significance
Ekadashi	11th tithi, both pakshas	Fasting day
Purnima	Full moon — tithi 15	Full moon observance
Amavasya	New moon — tithi 30	New moon observance
Sankashti Chaturthi	Krishna 4th	Ganesh observance
Vinayaka Chaturthi	Shukla 4th	Ganesh observance
Pradosh	Trayodashi (13th), both pakshas	Shiva observance

7. Festivals (v1 Scope)

Festival names are overlaid on the relevant date. Two categories are supported in v1:

7a. Tithi-Based Festivals

Festival	Lunar Month	Tithi
Chaitra Navratri (Day 1)	Chaitra	Shukla 1
Chaitra Navratri (Days 2-8)	Chaitra	Shukla 2-8
Ram Navami	Chaitra	Shukla 9
Hanuman Jayanti	Chaitra	Purnima (15)
Holi / Rangwali Holi	Chaitra	Krishna 1 (tithi 16)
Akshaya Tritiya	Vaishakha	Shukla 3
Buddha Purnima	Vaishakha	Purnima (15)
Nirjala Ekadashi	Jyeshtha	Shukla 11
Vat Savitri	Jyeshtha	Amavasya (30)
Rath Yatra	Ashadha	Shukla 2
Devshayani Ekadashi	Ashadha	Shukla 11
Guru Purnima	Ashadha	Purnima (15)
Nag Panchami	Shravana	Shukla 5
Raksha Bandhan	Shravana	Purnima (15)
Krishna Janmashtami	Bhadrapada	Krishna 8 (tithi 23)
Ganesh Chaturthi	Bhadrapada	Shukla 4
Anant Chaturdashi	Bhadrapada	Shukla 14
Sharad Navratri (Day 1)	Ashwina	Shukla 1
Sharad Navratri (Days 2-8)	Ashwina	Shukla 2-8
Vijayadashami / Dussehra	Ashwina	Shukla 10
Sharad Purnima	Ashwina	Purnima (15)
Dhanteras	Kartika	Krishna 13 (tithi 28)
Naraka Chaturdashi	Kartika	Krishna 14 (tithi 29)
Diwali	Kartika	Amavasya (30)

Govardhan Puja	Kartika	Shukla 1 (tithi 16)
Bhai Dooj	Kartika	Shukla 2 (tithi 17)
Chhath — Nahay Khay	Kartika	Shukla 4
Chhath — Kharna	Kartika	Shukla 5
Chhath — Sandhya Arghya	Kartika	Shukla 6
Chhath — Usha Arghya	Kartika	Shukla 7
Devutthana Ekadashi	Kartika	Shukla 11
Kartik Purnima	Kartika	Purnima (15)
Gita Jayanti	Margashirsha	Shukla 11
Vasant Panchami	Magha	Shukla 5
Maha Shivaratri	Phalguna	Krishna 14 (tithi 29)
Holika Dahan	Phalguna	Purnima (15)

7b. Solar Festivals

Festival	Condition
Makar Sankranti / Pongal	Sun enters Makara (sign 9)
Baisakhi / Vishu	Sun enters Mesha (sign 0)

Regional date variations, Onam (nakshatra-based detection), and multi-region festival differences are out of scope for v1.

8. Localization

Three display modes are user-selectable. All tithi names, paksha names, month names, and festival names must be available in all three.

Mode	Example / Note
English	"Shukla Pratipada"
Hindi — Latin script	"Shukla Pratipada" — may differ for festival names (e.g. "Deepavali" vs "Diwali")
Hindi — Devanagari	Rendered in Devanagari script (font support required)

9. Non-Functional Requirements

Requirement	Detail
Offline-first	No network required for any core function. Ephemeris files bundled.
Accuracy	Tithi transitions match published Drik Panchang sources to within 1 minute.
Performance	Main screen renders in under 200 ms on mid-range Android devices.
App size	Target under 30 MB installed (driven by ephemeris data).
Privacy	If GPS is used, location stays on-device. No analytics in v1.

10. Open Items — to Resolve Before / During Design Phase

Item	Status	Notes
App name	Resolved	Tithika
Visual theme	Resolved	Dark / night-sky background across all screens
Color palette	Pending	Shukla: #d9a441 · Krishna: #4a5fc1 · Accent: #c8421d — finalize in design
Typography	Pending	Especially Devanagari script support — evaluate Noto Sans Devanagari
Default location	Resolved	Set by user during first-launch onboarding (GPS or manual)
Onboarding fallback	Resolved	Ujjain fallback + persistent banner
Moon image animation	Pending	Static vs. animated phase transition